1 Station Andon Board: Layout Editor

Overview

Menu	Production facility/Resource management → Current information → Station Andon Board
Transaction code	stab
Function authorization	stab.editor

The editor is only available in English.

Purpose

The layout editor of the Station Andon Board allows you to design individual layouts visualizing any unit of the production line.

Integration

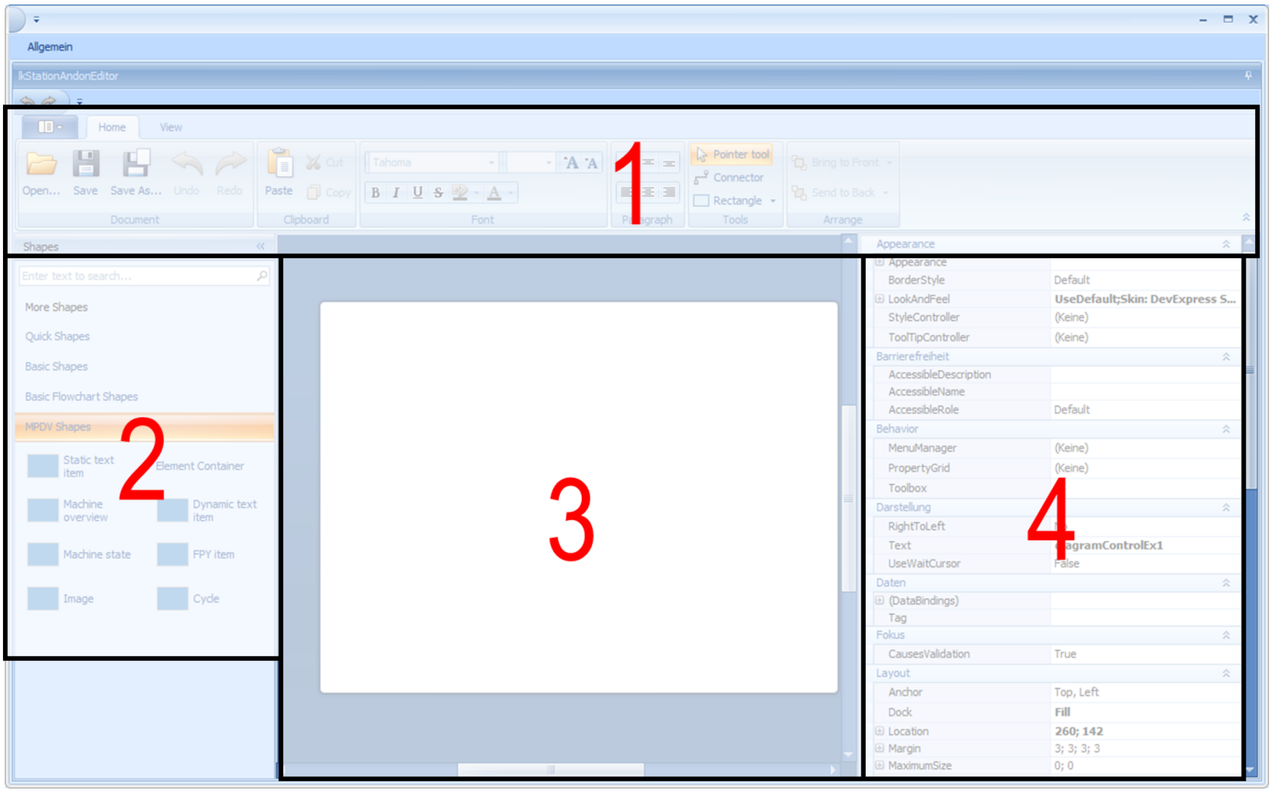
Use the messaging protocol MQTT to provide the MES-Weaver and MOC with data from the Dynamic MES-Weaver (DMW).

Requirements

You must configure MQTT communication in order to use the Station Andon Board. The required steps are described here.

General information on the layout editor

The layout editor has the following basic structure:



Pane (number)	Name	Purpose
1	Ribbon	The *ribbon* includes functions to open and store layouts. Additionally, you can use the given formatting functions to adjust the shapes (ready-made drawing objects) according to your requirements.
2	Shapes Panel	The *shapes panel* provides diverse shapes. You can drag and drop these shapes on the canvas.
3	Canvas	The *canvas* is the surface where you design the layout using shapes and connectors.
4	Properties Panel	The *properties panel* includes an object's properties. These properties vary depending on the object. The panel shows the properties of the selected object. You can change the properties to some extent.

Find further information on the layout editor here (in English).

MQTT connection

Use the MQTT messaging protocol to visualize DMW data in a timely manner. Use corresponding shapes to assign layout elements as well as topics and values.

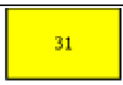
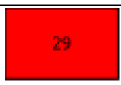
The MQTT connection is also enabled in the layout editor. The layout editor also shows changed MQTT items, which facilitates layout design.

MPDV shapes: general

There are special shapes for the Station Andon Board. You can find these shapes in the "MPDV Shapes" section of the shapes panel. Each shape has specific properties.

Cycle Shape

The *cycle shape* is a special shape to show the actual cycle of a machine including colored visualization. Maintain the following properties to use the *cycle shape:*

Property	**Meaning**	**Example**
Appearance → Content	MQTT communication identifies and assigns the content, name and/or field contents. MQTT communication overwrites any manually entered values.	
Appearance → MinValue_Green	The color *green* indicates values greater than this value.	
Appearance → MinValue_Yellow	The color *yellow* indicates values greater than this value and less than MinValue_Green.	
	The color *red* indicates values less than MinValue_Yellow.	
MQTT → Binding → Active	Enables MQTT communication for this shape. Must be set to "True" if the shape should show data received via MQTT.	
MQTT → Binding → Topic	Subscribed MQTT topic	

Property	Meaning	Example
MQTT → Binding → Value	Value to be visualized.	dmc.test.item

Dynamic Text Item Shape

The *dynamic text item shape* allows you to show any text from the data provided via MQTT. Maintain the following properties to use the *dynamic text item shape:*

Property	Meaning	Example
Appearance → Content	MQTT communication identifies and assigns the content, name and/or field contents. MQTT communication overwrites any manually entered values.	
MQTT → Binding → Active	Enables MQTT communication for this shape. Must be set to "True" if the shape should show data received via MQTT.	
MQTT → Binding → Topic	Subscribed MQTT topic	
MQTT → Binding → Value	Value to be visualized.	dmc.test.item

Element Container Shape

Use the *element container shape* to group single shapes. This facilitates positioning of related shapes on the canvas.

If you integrate MQTT-compatible shapes in the *element container shape,* you must use MPDV's *element container shape.* As this shape is compatible with MQTT communication.

FPY Item Shape

Like the *cycle shape,* the *FPY item shape* has been designed for the display and colored visualization of a specific value, i.e. in this case the *first pass yield* KPI (FPY). Data is displayed in percent.

Maintain the following properties to use the *FPY item shape:*

Property	Meaning	Example
Appearance → Content	MQTT communication identifies and assigns the content, name and/or field contents. MQTT communication overwrites any manually entered values.	
Appearance → MinValue_Green	The color *green* indicates values greater than this value.	96,7 %
Appearance → MinValue_Yellow	The color *yellow* indicates values greater than this value and less than MinValue_Green.	80,1 %
	The color *red* indicates values less than MinValue_Yellow.	73,3 %
MQTT → Binding → Active	Enables MQTT communication for this shape. Must be set to "True" if the shape should show data received via MQTT.	
MQTT → Binding → Topic	Subscribed MQTT topic	

Machine Overview Shape

The *machine overview shape* highlights the machine status in color. Unlike the *machine state shape,* the *machine overview shape* also includes the workplace name as a static value.

Maintain the following properties to use the *machine overview shape:*

Property	Meaning	Example
Appearance → Content	MQTT communication identifies and assigns the content, name and/or field contents.	
MQTT → Binding → Active	Enables MQTT communication for this shape.	

Property	Meaning	Example
	Must be set to "True" if the shape should show data received via MQTT.	
MQTT → Binding → Topic	Subscribed MQTT topic	

The application highlights machine statuses in the colors specified in the application Machines / Workplaces:

Light green	Status with RPA (resource performance account) 11 (usually “production")
Blue	Status with RPA 7 (usually "setup")
Red	Status = 30000 (usually "not assigned")
Gray	Status = 20000 or status with RPA 12 (usually "break“/“no shift“)
Yellow	Status < 10000 and RPA <> [11] and RPA <> 7

Machine State Shape

The *machine state shape* displays the machine status in color and as a text.

Property	Meaning	Example
Appearance → Content	MQTT communication identifies and assigns the content, name and/or field contents. MQTT communication overwrites any manually entered values.	
MQTT → Binding → Active	Enables MQTT communication for this shape. Must be set to "True" if the shape should show data received via MQTT.	
MQTT → Binding → Topic	Subscribed MQTT topic	

Static Text Item Shape

The *static text item shape* allows you to show static text in the layout. Maintain the following properties to use the *static text item shape:*

Property	Meaning	Example
Appearance → Content	Text to be displayed.	Hello

Specify background image

You can define a background image for the layout.

Property	Meaning	Example
Other → BackgroundImagePath	Path to the background image.	\\win2008-13\Hydra5\test_eke\Hallenlayout.png